

Parshvi Jain

parshvijain.vercel.app 📞 7452905310 ✉ parshvijain1508@gmail.com
🌐 linkedin.com/in/parshvi1508 🏠 github.com/parshvi1508

Education

ABES Engineering College, Ghaziabad

B.Tech in Computer Science (CGPA: 8.48 / 10)

2023 – Expected 2027

Ghaziabad, India

Indian Institute of Technology, Madras (Online)

BS in Data Science & Applications

2024 – Expected 2028

Chennai, India

Experience

Trurism (India)

Oct 2025 – Jan 2026

Backend Developer Intern

Remote

- * Built ML orchestration layer integrating model inference into booking and recommendation pipelines; integrated Razorpay HMAC-SHA256 webhooks and XMLAgency API for live flight and hotel data; shipped via Azure CI/CD with Docker and Unicorn.
- * Architected 20-module FastAPI backend with PostgreSQL, Alembic, and Redis; owned end-to-end design of JWT/RBAC auth, rate limiting, and multi-tenant white-label architecture supporting isolated B2B2C tenant operations across travel verticals.

MedEvidences AI (USA)

Apr 2025 – Aug 2025

Backend Developer Intern

Remote

- * Migrated Flask monolith to async FastAPI; rebuilt 30+ REST APIs with OAuth2 (Google, Microsoft), JWT authentication, and NPI middleware, achieving 25% latency reduction by eliminating blocking I/O across all database and auth calls.
- * Led zero-downtime PostgreSQL migration from Azure to AWS RDS using async SQLAlchemy; containerized with Docker; implemented dual-layer session and chat memory architecture for stateful AI workflows.

Projects

XAI Forensics – Transformer Decision Inspection Tool | Python, PyTorch, FastAPI, HuggingFace, LIME, Next.js, Docker | [GitHub](#) | [Demo](#)

- * Built a 3-panel forensic XAI system implementing LIME token attribution (300 perturbation samples), greedy counterfactual removal, and dual-model divergence scoring across DistilBERT-SST2 and Twitter-RoBERTa models deliberately chosen for domain mismatch so disagreement reflects genuine linguistic ambiguity, not noise; rejected attention weights as explanations per Jain & Wallace 2019.
- * Designed decoupled architecture with Dockerized FastAPI backend (4 endpoints, per-request latency tracking) and Next.js dashboard on Vercel; selected LIME over SHAP to preserve local fidelity on individual predictions, avoiding SHAP's global approximation assumptions for a per-request forensic use case.

Renewable Energy Forecasting: Pan-India LSTM | Python, TensorFlow, LSTM, scikit-learn, AWS S3 | [IEEE 2025](#)

- * Achieved 2.2% MAPE jointly forecasting energy consumption and renewable generation across 40+ Indian states, prior work addresses only one or the other, outperforming ARIMA (4.8%), linear regression (6.1%), and exponential smoothing (5.3%); RMSE = 0.15 GWh ($p < 0.01$). Presented at IEEE Conference 2025.
- * 4-layer LSTM (13K params), 60-day sliding windows, dropout-based uncertainty quantification (95% prediction intervals); automated ingestion of 96,360+ hourly records from AWS S3 via boto3; sub-1s CPU inference.

Data Analyst Agent API | Python, FastAPI, Gemini 2.5, Docker, OpenRouter | [GitHub](#)

- * Built an automated analysis API supporting 7 file formats via a 3-stage pipeline (Extract, Analyze, Format); designed semaphore-based concurrency (max 3 simultaneous requests) and sandboxed Docker execution with request-scoped UUID isolation to handle parallel workloads safely.
- * Implemented 4-attempt exponential backoff with Gemini 2.5-Flash primary and OpenRouter fallback to maintain uptime under model API instability; system continues serving requests even when primary model endpoint fails.

Technical Skills

Languages: Python, SQL

Frameworks: FastAPI, Flask, Next.js, Docker, Streamlit

AI/ML: PyTorch, scikit-learn, HuggingFace Transformers, TensorFlow, LIME, NLP

Cloud & Databases: AWS (S3, RDS), Azure, PostgreSQL, MySQL, Redis, Firebase, Git/GitHub

Concepts: System Design, REST APIs, Async Architecture, CI/CD, Multi-tenant Architecture

Achievements

IEEE Conference 2025: Presented research on LSTM-based pan-India renewable energy forecasting (2.2% MAPE). **NIT Agartala 2026:** Presented research on AI Interpretation by Women Across Generations. **Top 20 Worldwide:** Switch Case Contest, **ACM-W Global Newsletter:** First student chapter featured. **CodeChef:** Starters 189 – Rank 73, 188 – Rank 144, 176 – Rank 115.

Positions of Responsibility

Vice-Chair, ABES ACM-W (2025): Led technical and research initiatives; chapter recognized as Most Active in ACM India Region 2. **Panel Member, ACM India Region 2 Summit 2025. Blind Coding: Crown Edition:** Organized Codeforces contest with 100+ participants. **Cyberfeminism Sessions:** Delivered digital literacy workshops to 300+ girls.